Design and cloning of CRISPR guide RNAs

CRISPR endonucleases such as Cas9 and Cas12a enable targeted genome editing in a wide range of organisms. A short guide RNA (gRNA) directs the endonuclease to a genomic site based on sequence complementarity, typically requiring a short 2–6 bp specific protospacer adjacent motif (PAM).

Cas9 of *Streptococcus pyogenes* targets sites downstream of a 5'– –3' PAM and accepts a synthetic single guide RNA (sgRNA) that fuses the variable protospacer to the constant tracrRNA scaffold. Cas12a of *Acidaminococcus sp.*, by contrast, cleaves upstream of a 5'– !–3' PAM using a shorter crRNA without a tracrRNA, while Cas12a of *Lachnospiraceae sp.* also accepts 5'– "#!–3' PAMs.

Depending on the vector and Cas variant, guide RNAs may be cloned as individual expression units or as arrays of direct repeats and spacers. This protocol describes single (*Alternative A*) and pooled (*Alternative B*) guide RNA subcloning into U6 promoter–driven plasmids. The included examples are not exhaustive but illustrate distinct cloning architectures.

!"# $ %$&'

. !/0%12%34 '5!67%'%/ 829:41:'7'5!67%7 6 %#4!

▷ ;<%=<>%31?@!6 %A'/% 6#16B38 '/ /%"!5C%C1?1817(9#!4 35 #%4 :'!4%5 ?9 :8'5 /%6%147'6!/?/%3':'@8 %12%/ DB'8%4 :41#B35!16%14%7'? 517 6!3%3 88%8!6 /

▷ E '4%781 /+%/'2 5(%78'// /+%8'@%31'5

! " #$%&'(%)*+),)-

Procedures

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(1.) Review transcript variants of the target gene using Ensembl or UCSC Genome Browser. Download the relevant genomic sequence in FASTA format if required by your design platform.

- For gene knockout, choose early coding exons common to all isoforms.
- For in-frame tagging, retrieve genomic sequences within 50–70 bp of the start or stop codon.

*./ ' 0#12"*Make sure the target sequence is absent in the transgene to prevent re-cleavage after integration.

(2.) Design or select multiple candidate protospacer sequences using an online design tool.

!	"#$%&'!	('	)%!*+!',$ -.!	/!'+.-, &0!	1 2+0 3+&-	4!5!$!-*!
CRISPRDB	Knockout	SpCas9	Human, Mouse	Fixed library, Custom prediction	Post-hoc	[]
GeCKO	Knockout	SpCas9	Human, Mouse	Fixed library	Empirical	[]
E-CRISP	Knockout	SpCas9	Broad	Custom prediction	In silico	[]
PITCh designer	Tagging	User-specified	Broad	Custom prediction	In silico	[]
CCTop	Any	Many	Broad	Custom prediction	In silico	[]
CHOPCHOP	Any	Many	Broad	Custom prediction	In silico	[]

*+3'$2"*The table lists common CRISPR guide design tools. Some provide validated gRNA libraries like GeCKO, while others support custom protospacer prediction across species and Cason vosp O □ g